

RSA Digital Signature

Recall the RSA example that used a modulus of $n=(43)(59)=2537$, encryption exponent $e=11$, and decryption exponent $d=443$. We can use these parameters to create a digital signature for messages, but we sign messages using the private decryption exponent d .

Example:

Let's say that we send the message "quiz" to a recipient, and the recipient requires proof that it was actually us who sent it. Our modulus n and encryption exponent e are public, but our decryption exponent d is private (i.e. only we have knowledge of the factorization of n and hence the value of d).

We translate "quiz" into 4-digit blocks using $\mathbb{Z}_{26}$: quiz » 1620 0825

Then apply the private decryption exponent d to each block:

$$1620^{443} \bmod(2537) = 1610$$

$$0825^{443} \bmod(2537) = 1828$$

The signed message is "1610 1828"; only we could have signed it because only we know the value of d .

The recipient receives this signed message and verifies the signature by recovering the message using the public exponent $e=11$:

$$1610^{11} \bmod(2537) = 1620$$

$$1828^{11} \bmod(2537) = 0825$$

The recovered message is 1620 0825 » quiz. Note that the signed message "1610 1828" is not encrypted; anyone can use the public key $\{n, e\}$ to recover it.

In practice:

- The digital signature is applied to a hash of the message $h(M)$ to obtain a signed hash S .
- The pair (M, S) is transmitted by the sender, and can be further encrypted using RSA (or any other cryptosystem).
- The pair (M, S) can be verified by the receiver using the hashing algorithm to compute $h(M)$ and the public key of the sender $\{n, e\}$ to compute $h(M)$ from the signature S .
- A valid signature S can only be generated by someone who knows the private key $\{d\}$.

Exercises:

1. An RSA cryptosystem with public key $\{n=221, e=5\}$ is used to apply a digital signature to the message "199". Determine the decryption exponent d and sign the message.
2. You receive the message "jedi" from two people: Antilles and Ben.
Antilles sends the signature "2625 3901"
Ben sends the signature "2630 3074"
Both Antilles and Ben used an RSA cryptosystem with public key $\{n=3149, e=17\}$ to sign their messages using $\mathbb{Z}_{26}$ with a block size of 4-digits. Which signature is valid?

Answers:

1. $d = 77$

The signed message is "218"

2. Antilles: 1592 0893 (not valid)
Ben: 0904 0308 (valid)